

G-CAF

GovDigital Composable Architecture Framework

Volume 10

Worked Examples & Case Studies

v2.0

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Chapter 1

Examples Library Overview

This volume contains practical, worked examples demonstrating G-CAF concepts in action — the "show me how" companion to the framework documentation.

Example Categories

Category	Examples	Key Concepts
PBC Journeys	4 end-to-end lifecycle examples	Build/Buy journeys, gateway checklists, architecture diagrams
Agent Scenarios	3 AI agent implementations	Autonomy levels, guardrails, multi-agent orchestration
Classification Scenarios	3 decision-making examples	Sourcing trees, criticality scoring, orthogonal dimensions
Contract Evolution	3 schema change patterns	Additive changes, breaking migrations, event evolution
Case Studies	Template + contribution guide	Real-world adoption stories from entities

Quick Reference by PBC Type

PBC Type	Example
Application PBC	Custom Notification PBC (Build), Vendor Case Management (Buy)
Analytics PBC	Fraud Detection (Data Fabric consumption, no ownership)
Orchestration PBC	License Renewal (cross-PBC citizen journey, saga pattern)
Agent PBC	Document Processing (L1), Customer Service (L2), Multi-Agent (L3)

Quick Reference by Sourcing Strategy

Strategy	Example
Build	Custom Notification PBC, Fraud Detection PBC
Buy	Vendor Case Management (ServiceNow evaluation)
Outsource	Sourcing Decision Tree (identity verification scenario)

Chapter 2

PBC Journey: Custom Notification PBC

A citizen notification service sending alerts across SMS, email, push, and in-app. No existing PBC in the catalog — the team decides to build a shared service.

Phase 1: Initiation Gateway

Dimension	Selection	Rationale
PBC Type	Application PBC	Manages notification entities and templates
Criticality	Tier 2 (High)	Supports critical citizen communications
Sourcing	Build	Strategic capability, no COTS fit
Operational Model	Centralized	Shared service for all entities
Stability Tier	Stable	Core service, infrequent changes expected

Contract versions declared: Security 3.1, Data 2.0, Integration 2.2, NFR 1.5. Catalog search completed for "notification", "alert", "messaging" — no matches found. DDA Decision: Approved.

Phase 2: Design Gateway

Architecture: API Gateway → Notification Service (template management, channel routing, delivery tracking) → Channel adapters (SMS via Twilio, Email via SendGrid, Push via FCM/APNS, In-App via WebSocket). Event-driven: publishes notification.sent, notification.delivered, notification.failed events.

Interface contract: notification-api v1.0.0 (OpenAPI 3.0+). Schema registered in GovDX. Security: OAuth 2.0, PII masking in logs, encrypted at rest.

Phase 3: Go-Live Gateway

NFR validation passed (p95 latency < 200ms, 99.9% availability). Penetration test passed. Observability configured (metrics, logs, traces, dashboards). Support RACI documented. SLA published in PBC Catalog. Status updated to Active.

Chapter 3

PBC Journey: Vendor Case Management

Multiple entities need case management. Rather than building custom solutions, the decision is to procure ServiceNow and integrate it as a PBC within G-CAF governance.

Sourcing Decision

Is capability differentiating? NO (commodity). Does mature product exist? YES (ServiceNow, Salesforce, Dynamics). Can product meet 80%+ OOTB? YES. Decision: BUY.

Vendor Evaluation (DDA-led)

Criterion	ServiceNow	Salesforce	Weight
Functional fit	85%	78%	30%
Integration capability	APIs available	APIs available	20%
Security compliance	ISO 27001	SOC 2	20%
Schema evolution	Additive support	Additive support	15%
TCO (5 year)	\$2.1M	\$2.4M	15%
Final Score	84	76	—

Key risks identified: customization lock-in, upgrade complexity. Mitigations: limit customization, standard API facade (Anti-Corruption Layer).

Design Gateway Specifics for Buy

ACL (Anti-Corruption Layer) designed to translate between ServiceNow's internal model and G-CAF standard contracts. Interface contract published as case-api v1.0.0 in GovDX — consumers never interact directly with ServiceNow. Federated operational model: entities customize workflows within guardrails.

Chapter 4

PBC Journey: Analytics Fraud Detection

An Analytics PBC that analyzes transaction patterns across domains. Key distinction: it **consumes** from the Data Fabric but **does not own** any business entities.

Analytics PBC vs. Application PBC

Characteristic	Application PBC	Analytics PBC (This)
Data ownership	Owns entities	No ownership
Data source	Own database	Data Fabric consumption
Primary output	CRUD operations	Insights, scores, predictions
State	Persistent entities	Derived / computed

Data Fabric Integration

Consumes from 3 sources: Payment Processing PBC (real-time streaming transactions), Citizen Identity PBC (API query for verification results), License Management PBC (batch nightly applications). Data ownership: NONE — this is a defining characteristic of Analytics PBCs.

ML Architecture

Scoring API exposes fraud risk scores to consuming PBCs. ML pipeline: feature engineering from Data Fabric → model training (scheduled retraining) → model serving (real-time inference). ML Feature Store integration for reusable features. Model versioning and A/B testing support. Stability Tier: Evolving (ML models updated frequently).

Chapter 5

PBC Journey: License Renewal Orchestration

A citizen license renewal journey demonstrating cross-PBC orchestration: Login → Check Eligibility → Pay Fee → Receive License.

Step-by-Step Flow

Step	PBC	Operation	Event Published
1	Identity PBC	authenticateCitizen()	citizen.authenticated
2	License PBC	checkEligibility(citizenId)	license.eligibility.checked
3	License PBC	calculateFee(licenseId)	—
4	Payment PBC	initiatePayment(amount)	payment.initiated
5	Payment PBC	confirmPayment(reference)	payment.completed
6	License PBC	renewLicense(licenseId, paymentReference)	license.renewed
7	Notification PBC	sendConfirmation(citizenId)	notification.sent

Error Handling

Error Scenario	Handling	Compensation
Payment timeout	Retry with exponential backoff	Release fee hold
Eligibility fail	Show requirements to citizen	—
License system down	Queue for retry	Notify citizen of delay
Duplicate request	Idempotency check	Return existing result

Observability

Distributed trace spans the entire journey (12.3s total). Bottleneck identified: payment.process at 8.5s. Each PBC interaction captured as a child span with timing, enabling performance optimization.

Chapter 6

Agent Scenario: Document Processing Agent

An Agent PBC for high-volume permit application processing. Operates at **L1 (Semi-Autonomous)** — automates routine tasks, human decides edge cases.

Eight Governance Dimensions

Dimension	Design Decision
Identity	Service identity, no citizen impersonation
Autonomy	L1 — pre-approved actions only (classify, extract, validate)
Guardrails	Confidence threshold: 85%. Below → human review queue
Memory	Session-only, no persistent citizen memory across sessions
Interoperability	Consumes Fraud Detection PBC for risk scoring
Oversight	Human review queue for low-confidence classifications
Observability	Full trace logging of all classification decisions
Security	Zero Trust, no direct data access, API-only

Agent Architecture

Agent Core: Document Classifier (ML) → Field Extractor (ML) → Validation Engine (Rules). Guardrails Layer enforces confidence thresholds, action whitelists, PII masking, and escalation triggers. Human Review Queue handles items below 85% confidence. All decisions logged for audit.

Gateway requirement: Agent PBCs must declare all 8 governance dimensions at Initiation, plus standard contracts AND the Agent Governance Contract.

Chapter 7

Agent Scenario: Customer Service Agent

An **L2 (Supervised)** AI agent handling citizen inquiries across web chat, mobile app, and voice channels.

Architecture

Multi-channel (web chat, mobile, voice) → Agent Orchestration PBC (intent classification, context management, response generation, handoff coordination) → Domain PBCs (Citizen Profile, Service Catalog, Payment Gateway, Case Management).

Key Design Decisions

Decision	Implementation
Multi-Channel Consistency	Single agent model with channel-specific adapters
Graceful Degradation	Fallback to human agents when confidence below threshold
Privacy Compliance	No conversation retention beyond session
Audit Trail	Complete interaction logging for regulatory compliance
Autonomy Level	L2 — human handoff for sensitive actions (payments, PII changes)

Chapter 8

Agent Scenario: Multi-Agent Orchestration

Complex citizen requests requiring coordination between multiple specialized AI agents — Tax, Health, License, General — orchestrated by a central Orchestrator Agent.

Orchestration Patterns

Pattern	Use Case	Example
Sequential	Step-by-step processes	Tax filing → Payment → Confirmation
Parallel	Independent sub-queries	Check eligibility across multiple services simultaneously
Hierarchical	Complex multi-domain	Orchestrator delegates to domain-specific agents
Collaborative	Shared context	Agents contribute sections to a unified response

Autonomy: Orchestrator at L3, Domain Agents at L2. Memory: shared session context via Context Manager PBC. Observability: distributed tracing across all agent calls.

Conflict Resolution

Strategy	When Applied
Authoritative Source	Use data from the designated system of record
Recency	Prefer more recently updated information
Escalation	Flag for human review if agents provide unresolvable conflicting information
Transparency	Inform the citizen of uncertainty rather than guessing

Chapter 9

Classification: Sourcing Decision Tree

Step-by-step decision process: Is this strategic differentiator? → Do we have capability? → Is there a market solution? → BUILD / BUY / OUTSOURCE.

Decision Logic

- Strategic differentiator + have capability → **BUILD**
- Strategic differentiator + no capability → **OUTSOURCE** (build with partner, retain IP)
- Not strategic + market solution exists → **BUY**
- Not strategic + no market solution → **BUILD**

Evaluation Criteria Comparison

Factor	Build	Buy	Outsource
Time to Market	Slow	Fast	Medium
Total Cost (5yr)	High upfront	Recurring license	Variable
Customization	Full control	Limited by vendor	Negotiable
Vendor Lock-in	None	High risk	Medium risk
Maintenance	Internal team	Vendor handles	Partner handles
IP Ownership	Full ownership	None	Varies by contract

Example: Entity needs citizen identity verification. Strategic? YES. Capability? NO (no biometric expertise). Decision: OUTSOURCE — partner with specialist, retain IP, deploy as internal PBC.

Chapter 10

Classification: Criticality Tier Selection

Tier Definitions

Tier	SLA	RTO	RPO	Example
Tier 1	99.99%	15 min	0	Payment Gateway, Digital Identity
Tier 2	99.9%	1 hr	15 min	Case Management, Licensing
Tier 3	99.5%	4 hr	1 hr	Reporting, Analytics
Tier 4	99%	24 hr	4 hr	Archive, Batch Processing

Scoring Matrix

Factor	3 Points	2 Points	1 Point	0 Points
Revenue Impact	High	Medium	Low	None
Citizen Safety	Direct	—	Indirect	None
Regulatory Penalty	Severe	Moderate	Minor	—
Dependency Count	>10 PBCs	5–10 PBCs	1–4 PBCs	0 PBCs
Recovery Complexity	High	Medium	Low	—

Score 12–15 = Tier 1, Score 8–11 = Tier 2, Score 4–7 = Tier 3, Score 0–3 = Tier 4.

Example: License Renewal PBC

Factor	Assessment	Score
Revenue Impact	Medium — delays affect fee collection	2
Citizen Safety	None — administrative process	0
Regulatory Penalty	Moderate — compliance timelines	2
Dependency Count	6 PBCs depend on license status	2
Recovery Complexity	Medium — state restoration needed	2
Total		8 → Tier 2

Chapter 11

Classification: Independence

G-CAF uses orthogonal classification dimensions — each selected independently. This example shows how three different PBCs end up with very different combinations.

Example 1: Finance Platform PBC

Dimension	Selection	Rationale
PBC Type	Application	Manages financial entities
Criticality	Tier 1 (Critical)	Government financial operations
Sourcing	Buy (Oracle Fusion)	Commodity, mature product
Operational Model	Federated	Entity autonomy with central standards
Stability	Stable	Core finance, incremental changes

Implication: Critical + Buy + Federated = vendor must meet Tier 1 SLAs (99.99%), federated governance with central oversight, vendor contract includes strict support requirements, regular compliance audits.

Example 2: Legacy Integration PBC

Dimension	Selection	Rationale
PBC Type	Application	Connects to legacy mainframe
Criticality	Tier 3 (Standard)	Specialized, limited scope
Sourcing	Outsource	Complex, specialized skills needed
Operational Model	Distributed	Entity-specific, no sharing
Stability	Stable	Rarely changes after initial build

Implication: Standard + Outsource + Distributed = lower SLA acceptable (99.0%), vendor delivers and hands over, entity responsible for operations, minimal governance overhead.

Chapter 12

Contract Evolution: Three Patterns

Pattern 1: Additive Field Change (Non-Breaking)

Digital Identity PBC (Foundational tier, 47 consumers) needs to add preferredLanguage field. Analysis: additive, no removal, no type change → minor version bump (v2.3.0 → v2.4.0). DDA validates: truly additive, default behavior if absent, schema evolution rules followed. Automated compatibility check passes. Consumers notified but no action required (tolerant reader ignores unknown fields).

Pattern 2: Breaking Change Migration

Payment PBC changes amount from string to number. This is a breaking change requiring full governance process:

Phase	Timeline	Action
Announce v2 Beta	Week 0	RFC submission, consumers evaluate impact
v2 GA, v1 Deprecated	Week 4	v2 live, migration begins, dual-write period
v1 Warning	Week 12	Straggler consumers escalated to DDA
v1 End of Life	Week 24	Old version removed, all consumers on v2

Migration pattern: Phase 1 returns both formats (amount: 100.00 + amount_legacy: "100.00"), Phase 2 returns only new format, Phase 3 rejects old format.

Pattern 3: Event Schema Evolution

License PBC evolves the license.renewed event to add a structured fee object. Strategy: publish on both v1 and v2 topics simultaneously during transition. Legacy consumers continue on v1 topic. New consumers use v2 topic. v2 event includes deprecated feeAmount field for backward compatibility. CloudEvents envelope includes dataschema URL pointing to versioned JSON Schema.

Change Type	Compatibility	Strategy
Add optional field	Safe	Tolerant reader ignores
Add required field	Breaking	New event type or version
Remove field	Breaking	Deprecation period with parallel topics
Rename field	Breaking	Add new field + deprecate old
Change field type	Breaking	New event version with dual publishing

Chapter 13

Case Study Template & Contribution Guide

In community-driven adoption, entities trust entities more than they trust the EA Office. A case study answering "How did you handle legacy integration?" is worth more than a theoretical pattern document.

Case Study Categories

Category	Description	Example Topics
Adoption Journey	Entity's end-to-end G-CAF experience	Onboarding, change management, timeline, wins and setbacks
PBC Implementation	How a specific PBC was built/bought/composed	Contracted experience, contract compliance, technology choices
Integration Challenge	Cross-entity or legacy integration solved	Legacy wrapping, event-driven patterns, data migration
Governance in Practice	How governance worked (or didn't) in reality	Gateway friction, exception handling, DDA interactions
Reuse Story	How an entity reused another's PBC	Discovery, consumer onboarding, contract negotiation

What Makes a Good Case Study

Characteristic	What It Means
Honest	Includes what went wrong — "Share Your Scars" (G-CAF Way principle)
Specific	Names PBCs, contracts, timelines, and metrics; avoids vague generalities
Actionable	Ends with concrete advice another entity can act on immediately
Concise	2–4 pages maximum; busy architects won't read 20 pages
Quantified	Includes at least one measurable result (time saved, cost avoided, reuse achieved)

Template Structure

Every case study follows: **1. Challenge** (what was the problem?), **2. Approach** (what we did, what surprised us, where we got stuck), **3. Results** (measurable before/after metrics), **4. Lessons Learned** (what worked, what we'd do differently, advice), **5. Artefacts Produced** (reusable outputs for other entities).

Contribution Process

Write using template → Submit to Architecture Guild or EA Office → Guild reviews for quality → EA Office publishes → Present at CoP Meetup (optional). Contributing earns 20 points on the Contributor Ladder (L2+). Presenting adds 15 additional points.

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